



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 07

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Program:

```
def quick_sort(arr):  
    if len(arr) <= 1:  
        return arr  
  
    pivot = arr[-1]  
    left = []  
    right = []  
  
    for x in arr[:-1]:  
        if x <= pivot:  
            left.append(x)  
        else:  
            right.append(x)  
  
    return quick_sort(left) + [pivot] + quick_sort(right)  
  
arr = [10, 7, 8, 9, 1, 5]  
print("Sorted array:", quick_sort(arr))
```

Output:

The screenshot shows a Python IDE with a file named 'main.py'. The code defines a 'quick_sort' function that uses a recursive partitioning approach. It selects the last element as the pivot, partitions the array into elements less than or equal to the pivot and elements greater than the pivot, and then recursively sorts both sub-arrays. The array [10, 7, 8, 9, 1, 5] is passed to the function, and the output is printed as 'Sorted array: [1, 5, 7, 8, 9, 10]'. The IDE interface includes a 'Visualize' button and a 'Run' button with a play icon.

```
main.py Visualize Run ▶ Output  
1 def quick_sort(arr):  
2     if len(arr) <= 1:  
3         return arr  
4  
5     pivot = arr[-1]  
6     left = []  
7     right = []  
8  
9     for x in arr[:-1]:  
10        if x <= pivot:  
11            left.append(x)  
12        else:  
13            right.append(x)  
14  
15        return quick_sort(left) + [pivot] + quick_sort(right)  
16  
17 arr = [10, 7, 8, 9, 1, 5]  
18  
19  
20 print("Sorted array:", quick_sort(arr))
```

Sorted array: [1, 5, 7, 8, 9, 10]